

- 3 Two progressive water waves X and Y travel along a straight line from point A to point B. The variation of displacement of the waves with distance from A at an instant in time is shown in Fig. 3.1.

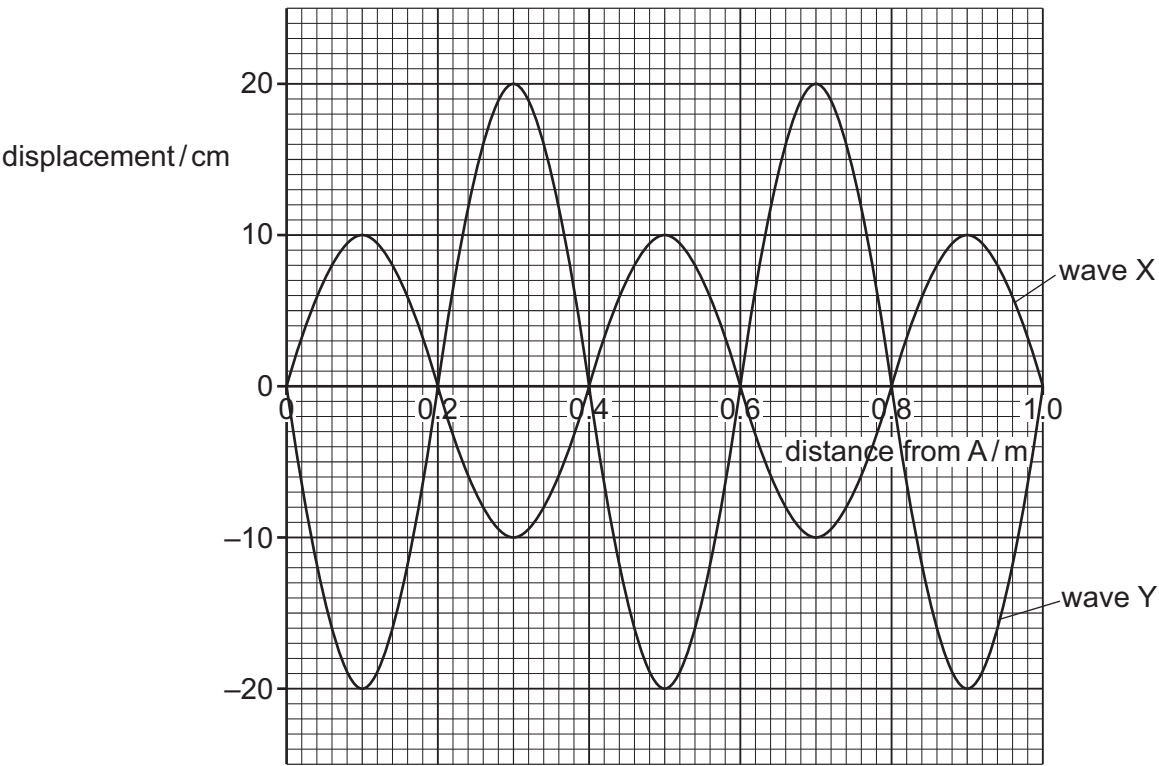


Fig. 3.1

- (a) State the amplitude of wave X.

[1]
- (b) Both waves have frequency 16 Hz.

(i) Determine the speed of wave X.

[2]

(ii) State and explain whether X and Y are coherent.

[1]

(c) Wave X and wave Y superpose to form a resultant wave.
- On Fig. 3.2, sketch the variation of displacement of the resultant wave with distance from A at the instant of time shown in Fig. 3.1.
- This is a blank graph for sketching the resultant wave. The axes are identical to Fig. 3.1: the vertical axis is 'displacement / cm' from -20 to 20, and the horizontal axis is 'distance from A / m' from 0 to 1.0. The grid consists of major lines every 10 units on the vertical axis and every 0.2 units on the horizontal axis, with minor lines every 2 units and every 0.02 units respectively.
- Fig. 3.2
- (d) The intensity of wave X is I_X . The intensity of wave Y is I_Y .

Use Fig. 3.1 to determine the ratio $\frac{I_X}{I_Y}$.
- [Total: 8]